Ulver Spangsberg Rouw

MD, PhD | Vascular Surgery | Clinical AI & Medical Writing



Contact

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Interests

Outside work, most of my interests still involve deliberate practice, preferably with my hands. I spend much of my spare time with my wife and two sons, and I also climb, paint miniatures, and occasionally pretend that former fencing reflexes are still clinically useful.
I am a former regional fencing champion and previously reached the "1000-pound club" in strength training. These days, climbing has taken over: technical, physical, mildly obsessive, and therefore not entirely unlike surgery.

References

Jonas Peter Eiberg, MD, PhD
Senior Consultant Surgeon
Vascular Surgery Department
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Profile

Clinician-scientist in vascular surgery with an MD, PhD and specialist training at Rigshospitalet. I combine surgical experience with clinical research, medical AI, ultrasound technology and applied statistics, and have personally performed more than 3,500 vascular ultrasound examinations.

My PhD focused on ultrasound-based biomechanical assessment of abdominal aortic aneurysms, including automated image analysis, strain-pattern modelling, and machine-learning prediction of aneurysm growth.

Alongside my specialist training, I am interested in focused part-time consultancy work at the interface between medicine, research and technology. This could include medical writing, project planning, AI model training and validation, and statistics.

Education

July 2018 — **MD (Cand. Med.)**
· University of Copenhagen

March 2026 — **PhD**
· University of Copenhagen
Defended PhD thesis entitled ***Quantification of abdominal aortic aneurysm biomechanical properties with ultrasound***

Employment

Dec 2025–Apr 2026 — **Clinical AI Consultant**
Philips Ultrasound LLC · Remote

Nov 2025–Sep 2030 — **Specialist Registrar in Vascular Surgery**
Department of Vascular Surgery · Rigshospitalet

Jun 2022–Oct 2025 — **PhD Student**
Vascular Research Unit · Rigshospitalet

Jun 2023–May 2024 — **Varicose Vein Assistant**
Dansk Venecenter · Nordic Health Group

Nov 2021–May 2022 — **Researcher**
Vascular Research Unit · Rigshospitalet

Oct 2020–Oct 2021 — **Resident in Vascular Surgery**
Department of Vascular Surgery · Rigshospitalet

Oct 2019–Sep 2020 — **Postgraduate Internship (KBU)**
Orthopedics & General practice · Nykøbing F. Hospital

Aug 2018–Sep 2019 — **Unclassified Position in Vascular Surgery**
Department of Vascular Surgery · Rigshospitalet

Research Output

18 peer-reviewed publications, including 8 first-author and 2 last-author papers, with recent work in the *European Journal of Vascular and Endovascular Surgery*. Additional manuscripts are in review or in preparation, including work on automated ultrasound surveillance, strain-pattern analysis and machine-learning prediction of aneurysm growth.

Selected first-author work

2026 **Rouw US** et al. Validation of Automated Strain Calculation Initialised by Deep Learning Segmentation for Abdominal Aortic Aneurysm Ultrasound. *Eur J Vasc Endovasc Surg*.

2026 **Rouw US** et al. Ultrasound strain patterns are associated with abdominal aortic aneurysm growth. *Eur J Vasc Endovasc Surg*.

2025 **Lorenzen US** et al. Strain patterns with ultrasound for assessment of abdominal aortic aneurysm vessel wall biomechanics. *Ultrasound Med Biol*.

2024 **Lorenzen US** et al. Abdominal aortic aneurysm volume growth patterns with three-dimensional ultrasound. *Int Angiol*.

2021 **Lorenzen US** et al. The short-term predictive value of vessel wall stiffness on abdominal aortic aneurysm growth. *Ann Vasc Surg*.

2019 **Lorenzen US** et al. Coronary artery bypass surgery plus medical therapy vs medical therapy alone for ischaemic heart disease: protocol for a systematic review with meta-analysis and TSA. *Syst Rev*.

Selected Presentations & Awards

2025 Data-Driven Surgery, Copenhagen and Danish Society for Vascular Surgery, Aalborg — oral presentations on ML-aided AAA surveillance.

2024 Annual Meeting, Danish Society for Vascular Surgery, Copenhagen — oral, **winner of the abstract competition**.

2021 ESVS Annual Meeting, Rotterdam — **prize oral presentation** on aortic biomarkers and aneurysm growth.

2020–2024 ESVS Annual Meetings: Krakow 2024 (poster), Belfast 2023, Rome 2022 (supervised presenter), Rotterdam 2021 (×2), online 2020 (e-poster).

Clinical & Surgical Experience

Almost 500 documented vascular procedures across approximately 5 years at Rigshospitalet's Department of Vascular Surgery and the Nordic Health Group. Experience spans open and endovascular aortic repair, peripheral bypass and endarterectomy, AV-fistula surgery, superficial venous procedures and vascular trauma or revision surgery.

This clinical background is central to my consultancy profile: I understand how medical technologies are used in real workflows, where validation can fail, and what clinicians need before trusting a new tool.

Supervision, Teaching & Leadership

Primary supervisor for medical students, junior doctors, and an international PhD fellow, with projects covering automated 3D ultrasound, aortic aneurysm strain-pattern analysis, biomarker-based growth prediction and surveillance after EVAR. Several supervised projects have led to publications, accepted manuscripts or conference presentations.

Instructor on the Basic Course in Vascular Ultrasound and former Resident Education Coordinator at the Department of Vascular Surgery, Rigshospitalet. Completed 38.2 ECTS of PhD-level coursework, including R programming, Bayesian methods, survival analysis, repeated measurements and causal inference.

Collaborations

2025 Section of Biostatistics, University of Copenhagen: functional data analysis with Assoc. Prof. Andreas Kryger Jensen.

since 2025 Leiden University Medical Center × Vascular Research Unit, Rigshospitalet.

since 2021 Philips Health Technology Innovation × Vascular Research Unit, Rigshospitalet.